

Lesson 09 Demo 02

Managing Cosmos DB

Objective: To create a Cosmos DB account, explore deployment and configuration options, and add data to verify its functionality

Tools required: Azure account with administrator access

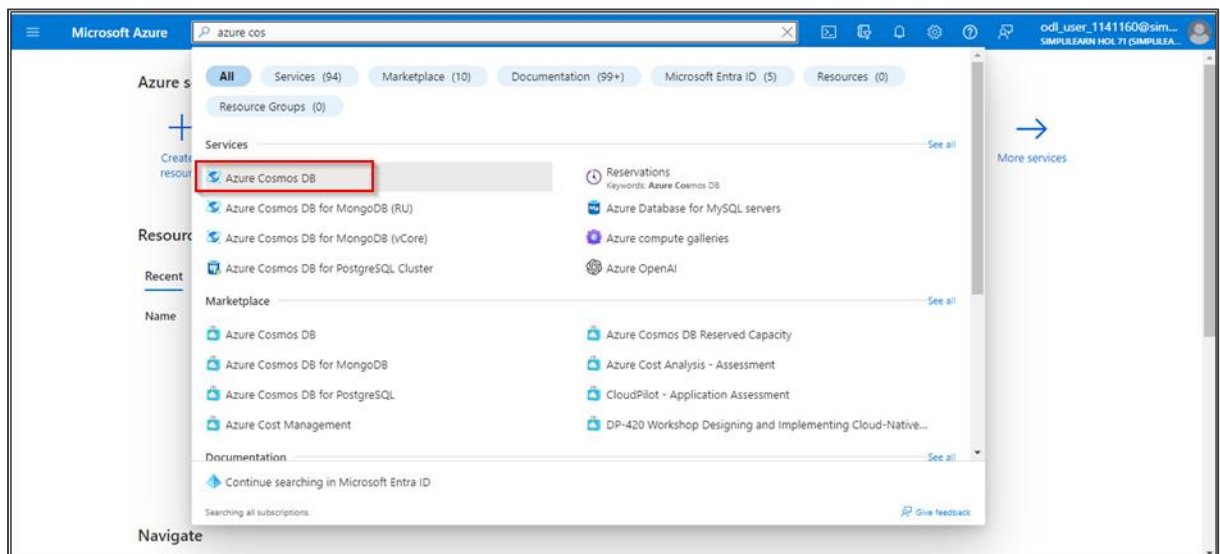
Prerequisites: None

Steps to be followed:

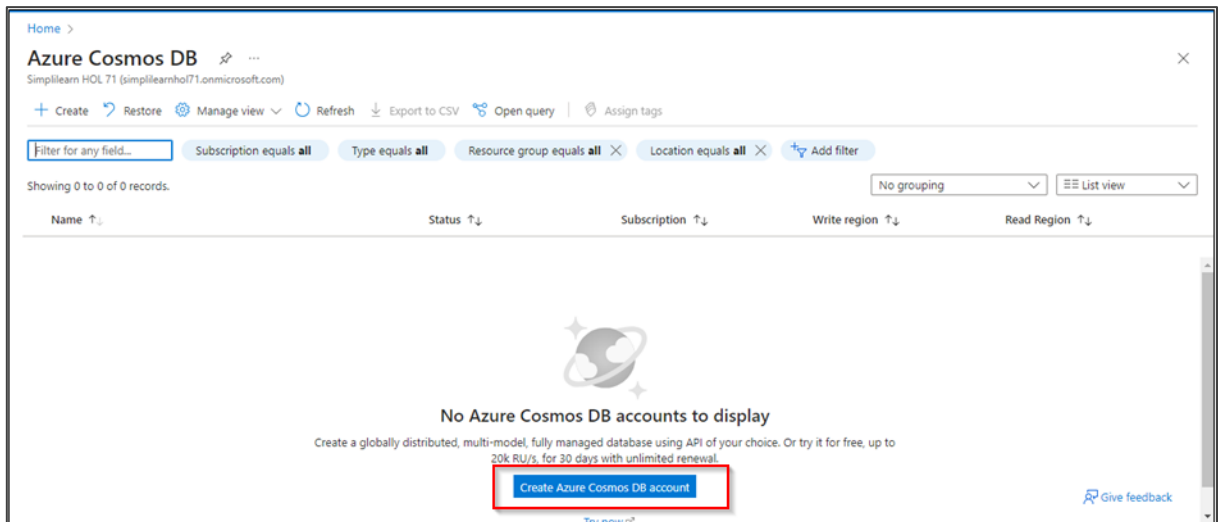
1. Create a Cosmos DB account
2. Create a database and container
3. Add data to the cosmos DB database
4. Query the data using SQL APIs

Step 1: Create a Cosmos DB account

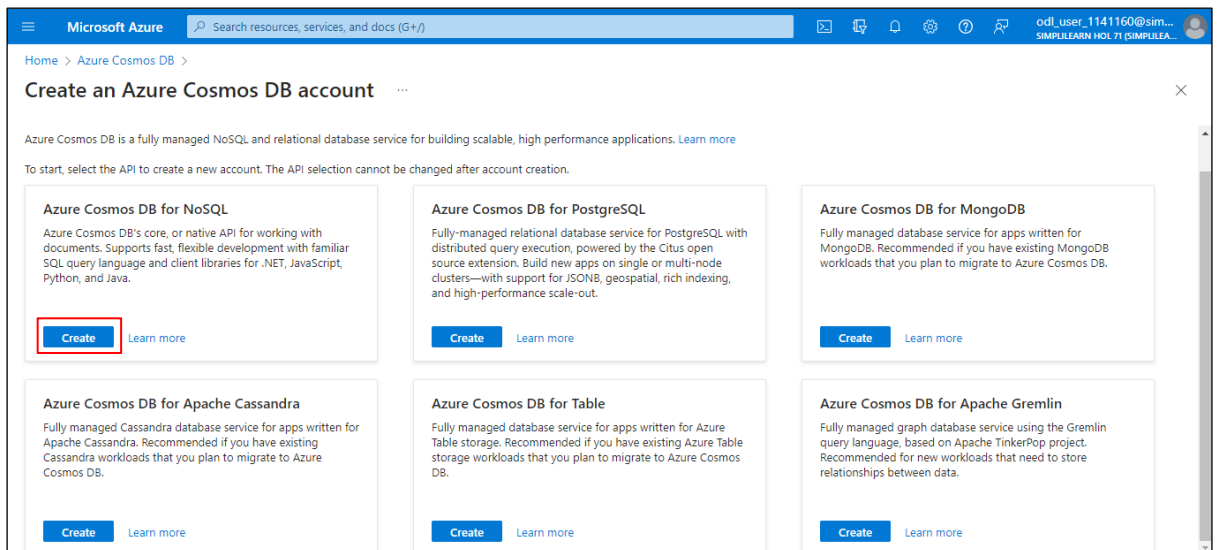
1.1 Log in to your Azure account; search for and select **Azure Cosmos DB**



1.2 In the Azure Cosmos DB page, click on Create Azure Cosmos DB account



1.3 In the Create an Azure Cosmos DB account page, click on Create under Azure Cosmos DB for NoSQL



1.4 Navigate to the **Basics** tab. Under **Project Details**, click on **Create new** to create a new resource group and provide information as given in the screenshot below. Click on **Review + create**.

Microsoft Azure

Home > Azure Cosmos DB >

Create Azure Cosmos DB Account - Azure Cosmos DB for NoSQL

Basics Global Distribution Networking Backup Policy Encryption Tags Review + create

Azure Cosmos DB is a fully managed NoSQL and relational database service for building scalable, high performance applications. [Try it for free](#), for 30 days with unlimited renewals. Go to production starting at \$24/month per database, multiple containers included. [Learn more](#)

Project Details
Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

Resource Group *
[Create new](#)

Instance Details

Account Name *

Location *

Capacity mode ☐ Provisioned throughput ☒ Serverless
[Learn more about capacity mode](#)

[Review + create](#) [Previous](#) [Next: Global Distribution](#)

1.5 After the validation is successful, click on **Create**

Microsoft Azure

Home > Azure Cosmos DB >

Create Azure Cosmos DB Account - Azure Cosmos DB for NoSQL

Basics Global Distribution Networking Backup Policy Encryption Tags Review + create

Validation Success

Creation Time
Estimated Account Creation Time (in minutes) 2
The estimated creation time is calculated based on the location you have selected

Basics

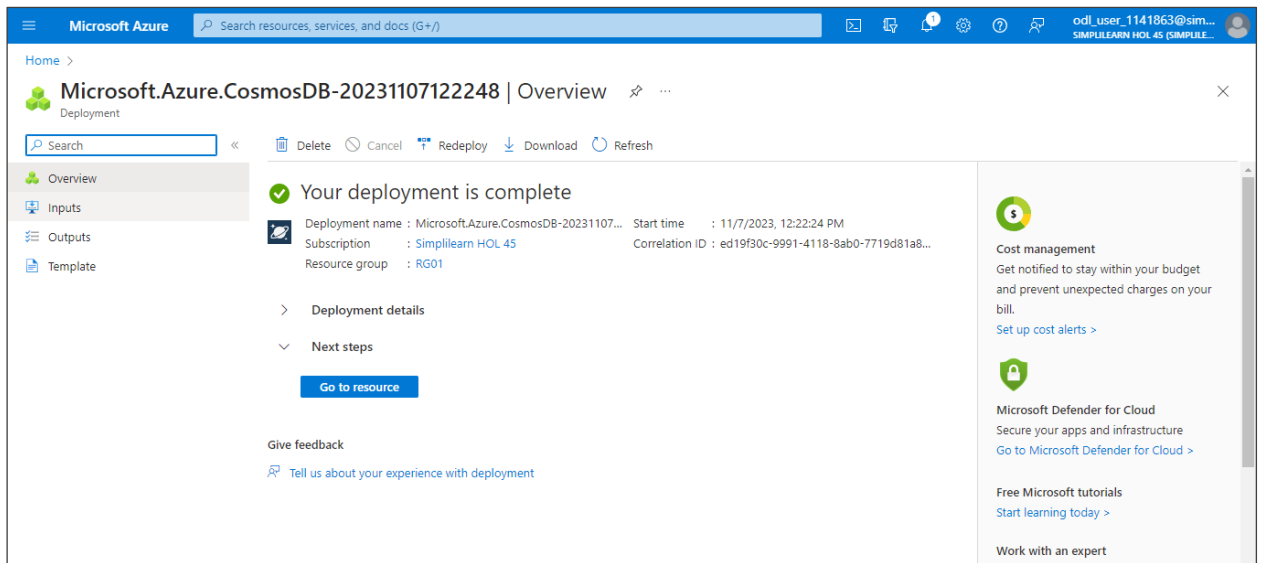
Subscription	Simplilearn HDL T1
Resource Group	(new) RG01
Location	West US
Account Name	(new) mycosmos24
API	Azure Cosmos DB for NoSQL
Capacity mode	Serverless

Backup Policy

Backup policy	Periodic
Backup storage redundancy	Geo-redundant backup storage

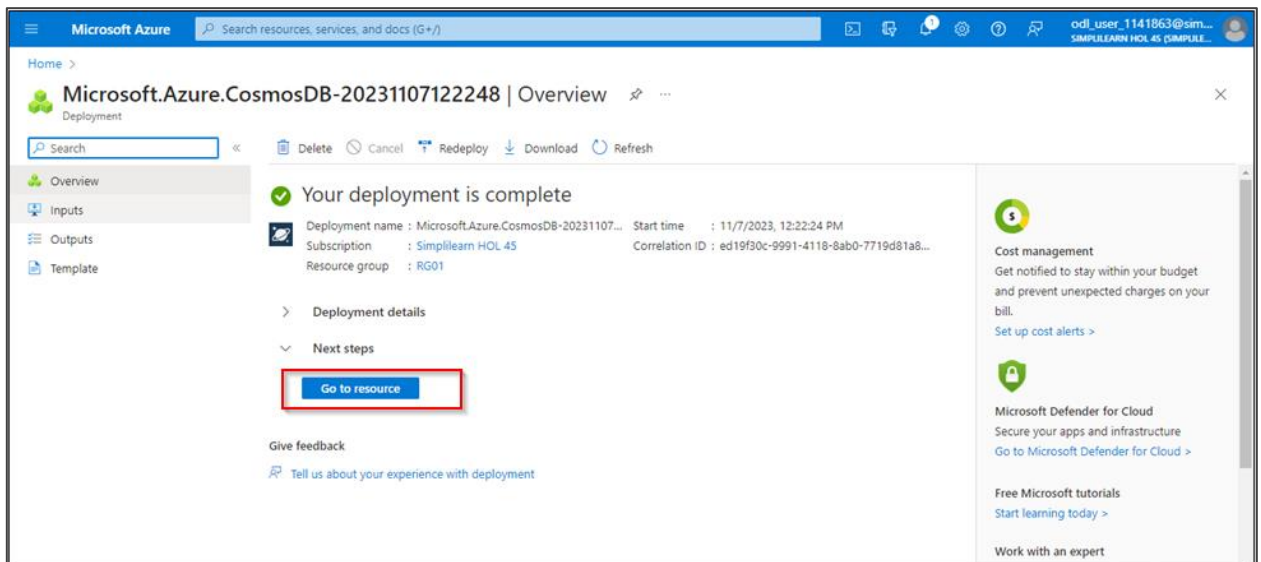
[Create](#) [Previous](#) [Next](#) [Download a template for automation](#)

The deployment completion screen appears.

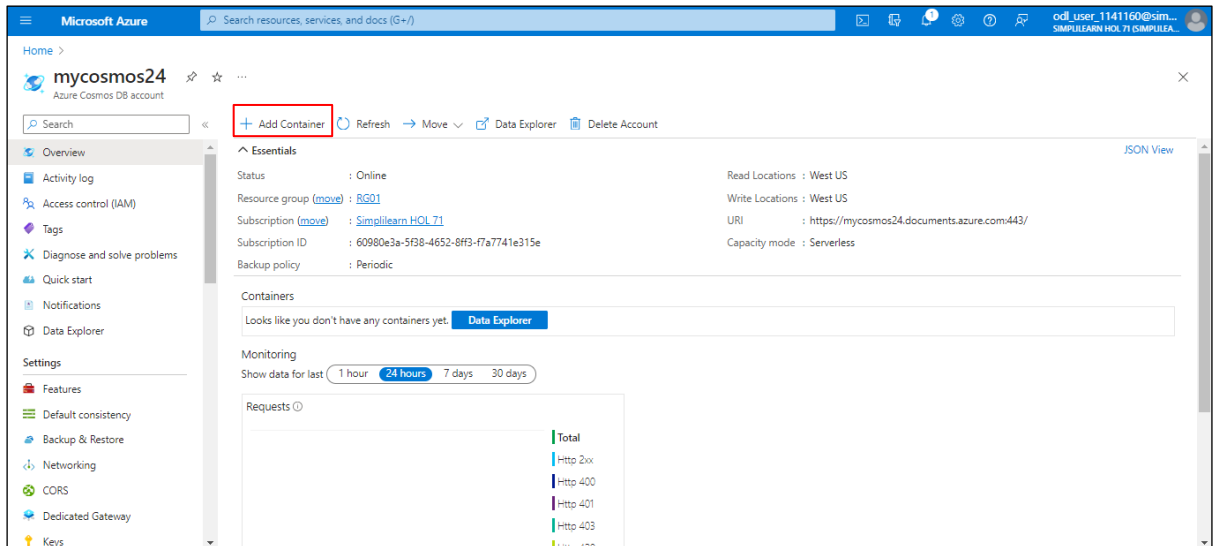


Step 2: Create a database and container

2.1 Click on **Go to resource**



2.2 In the newly created Azure Cosmos DB account (**mycosmos24**) page, click on **Add Container**

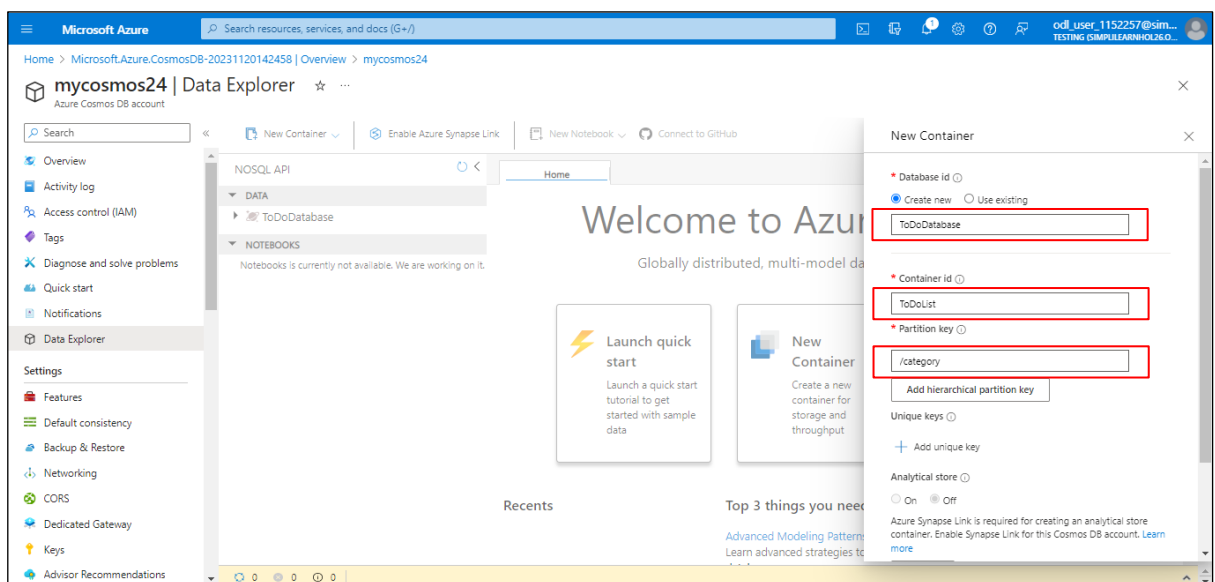


2.3 Provide values for all fields as mentioned below:

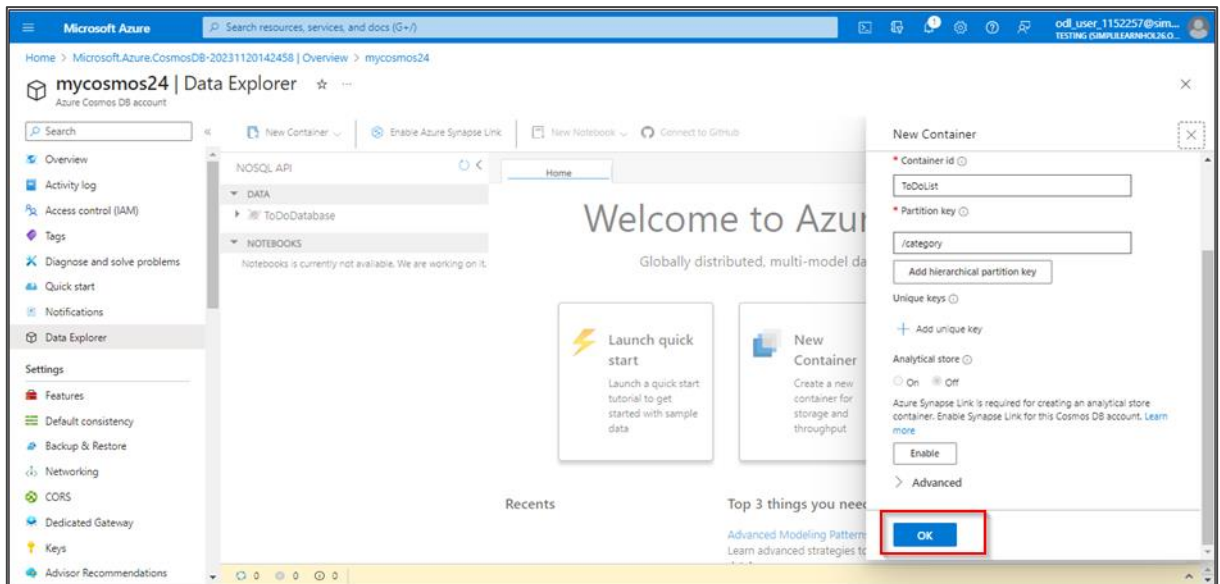
Database id: ToDoDatabase

Container id: ToDoList

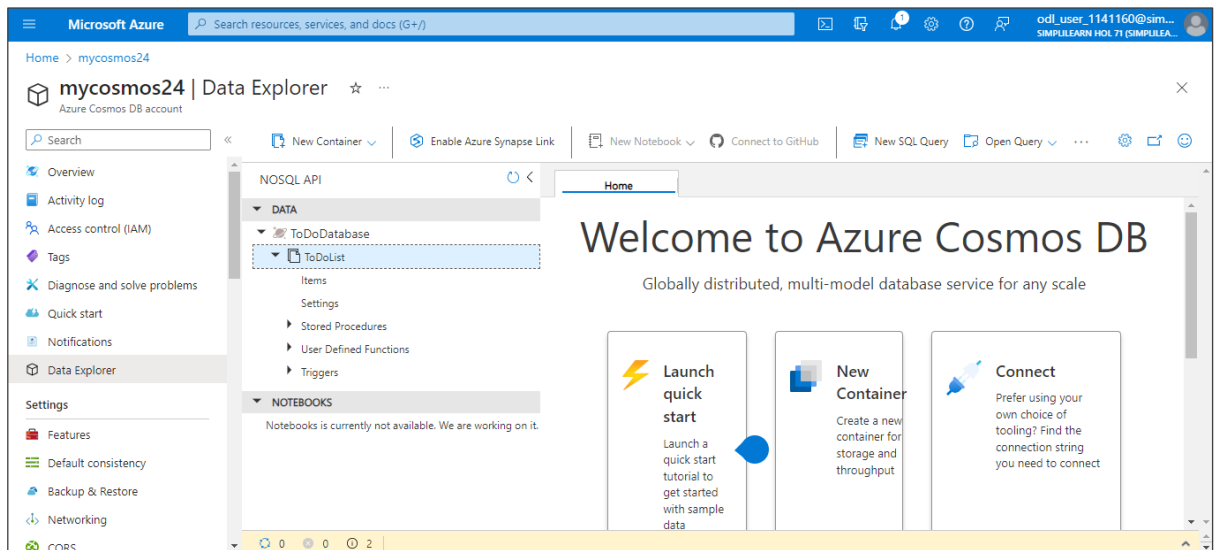
Partition key: /category



2.4 After entering the details, scroll down and click on **OK**

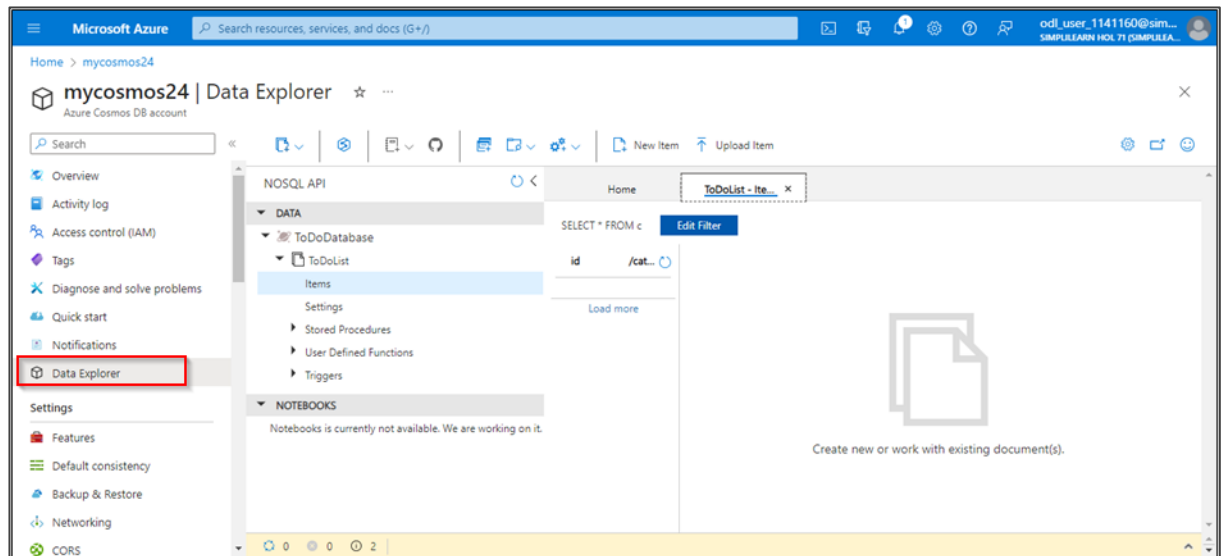


Once the database is created, the hierarchy should look like the one below:

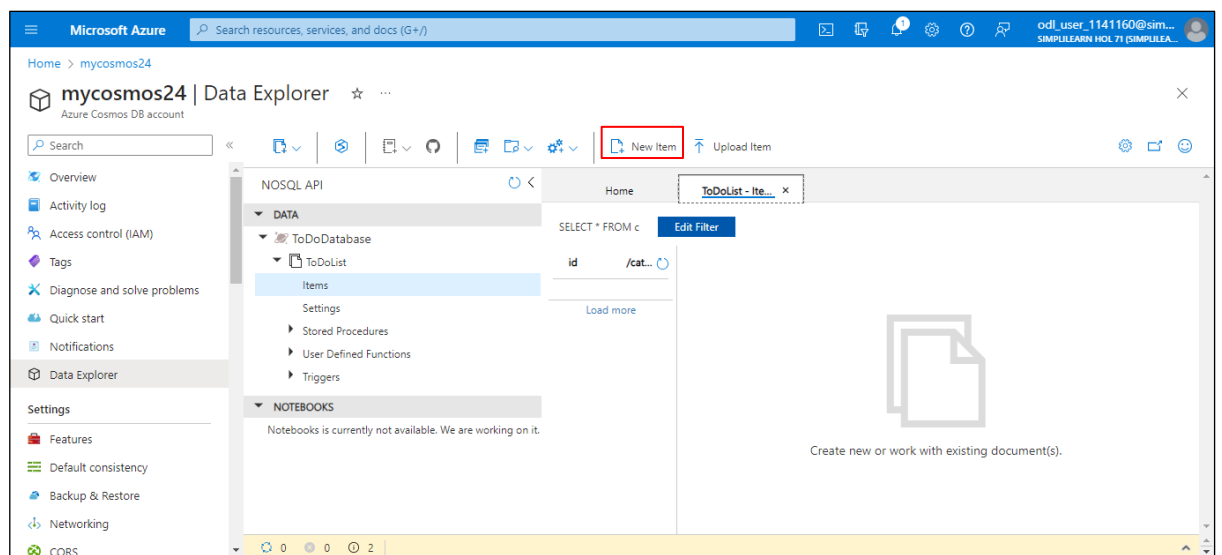


Step 3: Add data to the Cosmos DB database

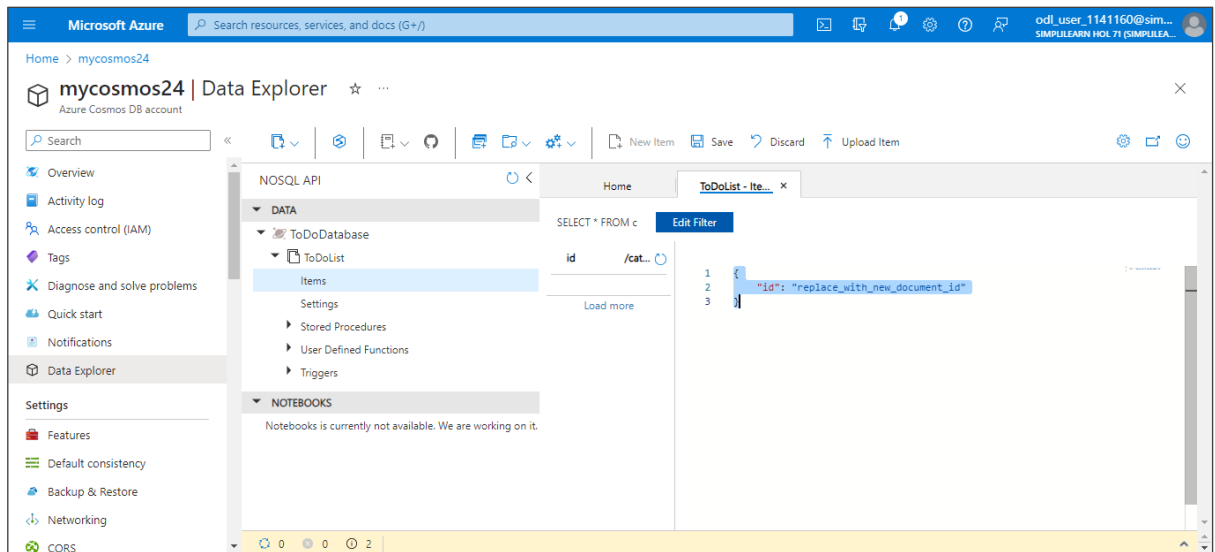
3.1 In the **Cosmos DB** account page, click on **Data Explorer** on the left side panel



3.2 Expand **ToDoDatabase** and **ToDoList**. Then, click on **Items** and select **New Item**.

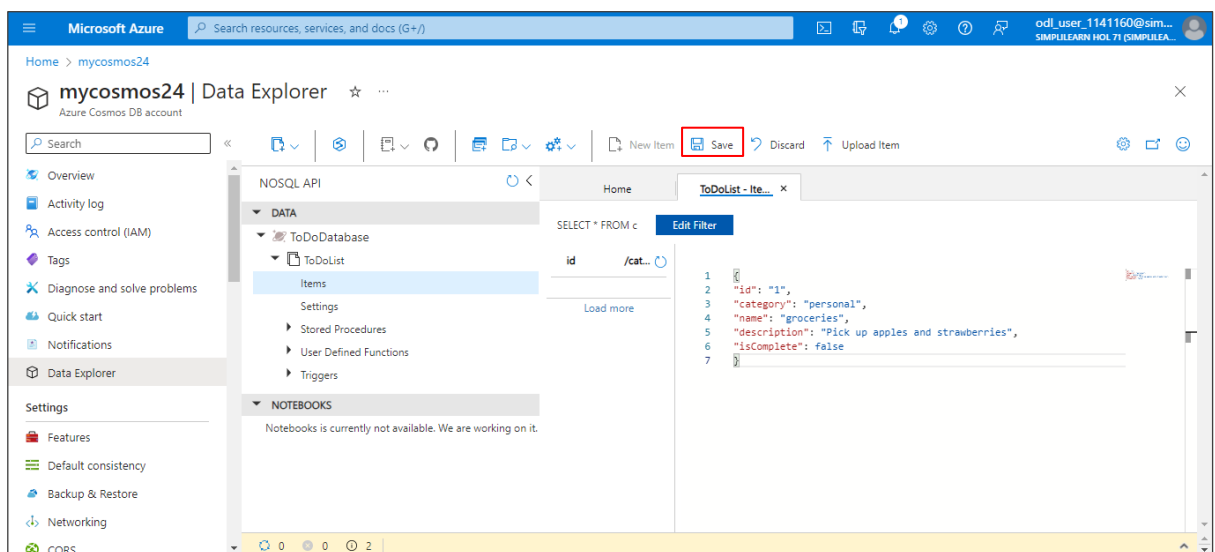


3.3 Select the default data in the field below the **Edit Filter** button



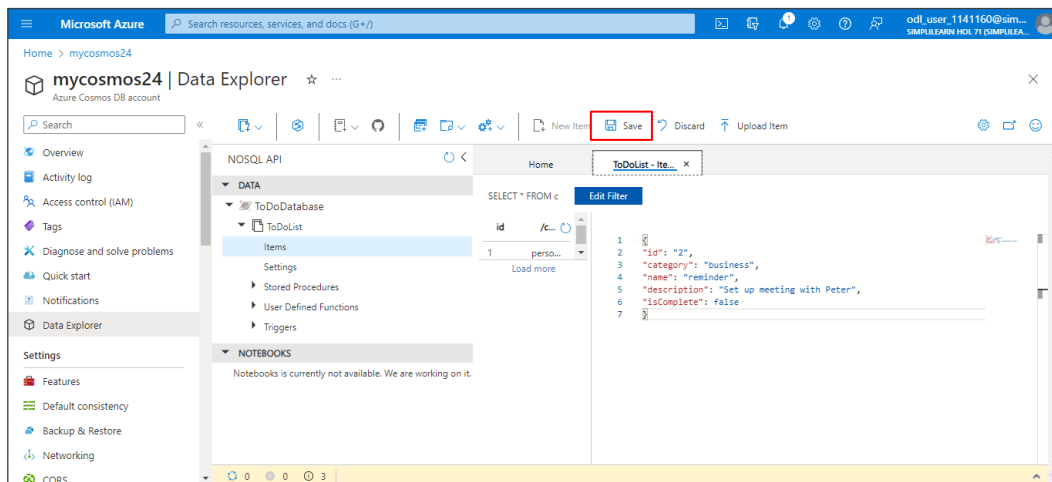
3.4 Replace the selected data with the below code and click on **Save**:

```
{
  "id": "1",
  "category": "personal",
  "name": "groceries",
  "description": "Pick up apples and strawberries",
  "isComplete": false
}
```



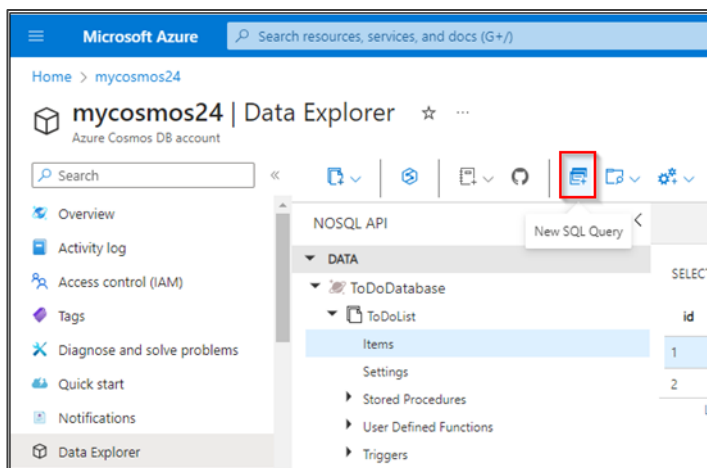
3.5 Similarly, add the next item and click on **Save**

```
{
  "id": "2",
  "category": "business",
  "name": "reminder",
  "description": "Set up meeting with Peter",
  "isComplete": false
}
```



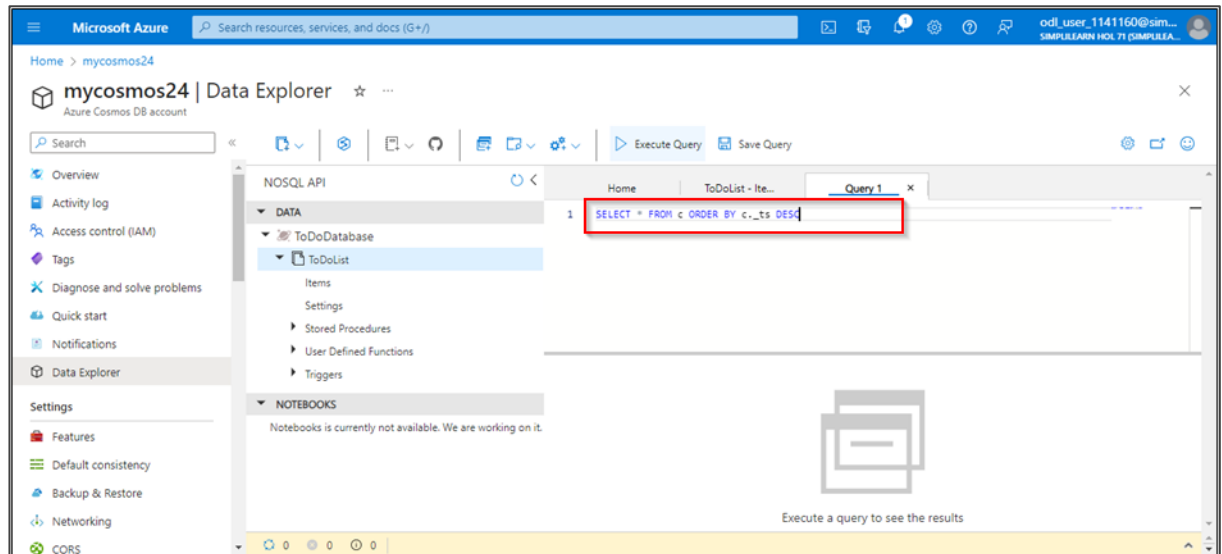
Step 4: Query the data using SQL APIs

4.1 To run a query, click on the icon as shown below and select **New SQL Query**

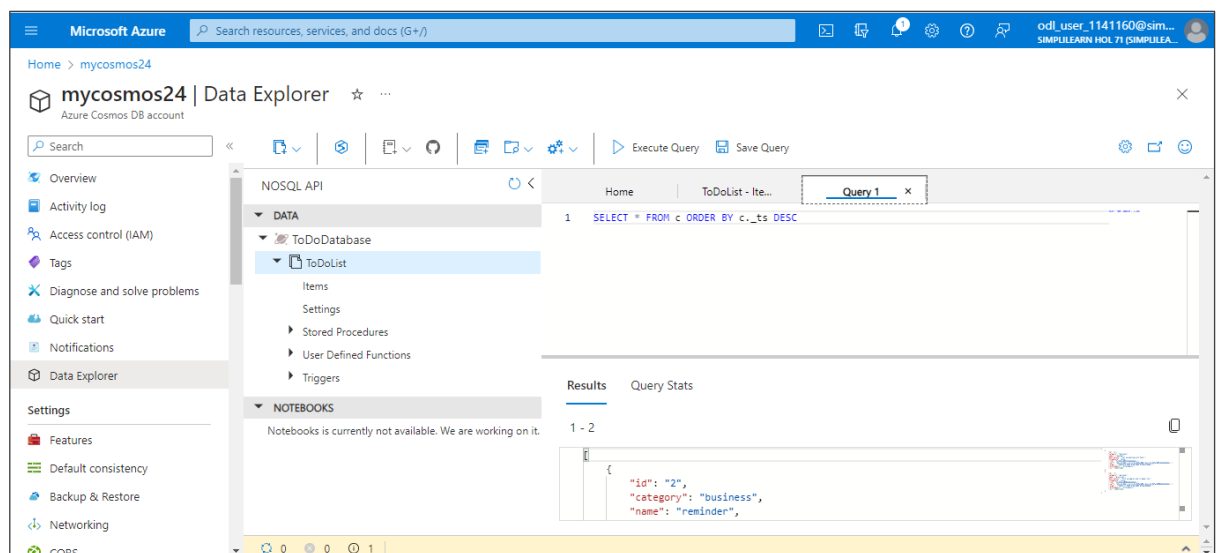


4.2 Run the below query and click on **Execute Query**

SELECT * FROM c ORDER BY c._ts DESC



The modified query displays the documents in descending order based on their time stamp, so now your second document is listed first.



By following the above steps, you have successfully created a Cosmos DB account, identified different options for deploying and configuring it, and added some data to check its functionality.